SECTION 1: OVERVIEW OF SURVEY

Thank you for participating in this survey. Your expertise is invaluable to our research on the sustainable conservation of Nepal's unique historic bridges.

This questionnaire is designed to gather your informed perspectives on the **relative importance** of various factors that should guide decisions about protecting and managing these vital heritage structures. We are employing a widely recognized decision-making tool called the **Analytic Hierarchy Process (AHP)** to systematically incorporate expert judgment.

What You Will Be Doing:

You will be presented with a series of questions where you compare two different factors (e.g., "Material degradation" vs. "Foundation vulnerability"). For each comparison, you will indicate which factor is more important, and by how much, using a specific numerical scale (mentioned in later part).

The questionnaire is divided into four main sections:

1. **Overview of Survey / Introduction:** This is the initial part that explains what the survey is about, why it's being conducted, how the AHP scale works, and how your responses will be used in the research. It sets the context for the entire questionnaire.
2. **Structural Indicators:** You will compare different aspects of a bridge's physical condition and engineering integrity.
3. **Cultural Indicators:** You will compare various elements that define a bridge's historical, architectural, and community value.
4. **Overall Category Importance:** You will determine the relative importance of "Structural" considerations versus "Cultural" considerations when making a comprehensive conservation decision.

Understanding the Scale for Your Responses

For each question, you will be comparing two items and deciding which one is more important, and by how much. You will use a scale from 1 to 9, where:

- **1: Equal importance** (Both items are equally important)
- **3: Moderate importance** (The first item is moderately more important than the second)
- **5: Strong importance** (The first item is strongly more important than the second)
- **7: Very strong importance** (The first item is very strongly more important than the second)
- **9: Extreme importance** (The first item is overwhelmingly more important than the second)

You can also use **2, 4, 6, 8** for intermediate values.

What if the SECOND item is more important?
If the second item in the comparison is more important, you should select the **reciprocal**

value (e.g., if the second item is moderately more important, select **1/3,** Similarly, if second item is strongly more important than the first one enter **1/7**).

How Your Responses Will Be Used:

Your responses will be directly used to **calculate numerical 'weights'** for each indicator and for the overall structural and cultural categories. These weights reflect your collective expert judgment on what matters most in evaluating a historic bridge.

Specifically:

For example, if you consistently rate "**Material degradation**" as 'strongly more important' than "**Joint condition,**" this will result in a higher calculated weight for "**Material degradation**." Similarly, If you deem "**Structural**" aspects '**equally important**' to "**Cultural**" aspects overall, their final weights in the framework will reflect this balance.

* Indicates required question

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SECTION 2: RELATIVE IMPORTANCE OF STRUCTURAL INDICATORS

In this section, please compare pairs of **structural indicators** to determine their relative importance in assessing the physical condition and integrity of historic bridges.

Structural Indicators:

- Material degradation (e.g., decay of timber, cracking of stone, corrosion)
- Foundation/scour vulnerability (e.g., erosion around foundations, undermining)
- Structural deformation (e.g., warping, sagging, misalignment)
- Joint condition (e.g., looseness, corrosion, missing elements in connections)
- Deck integrity (e.g., damage to walking surface or deck beams)
- Live load response (e.g., observable behavior under pedestrian load, excessive bounce)
- Environmental exposure (e.g., proximity to riverbank erosion, flood debris, landslides)

Understanding the Scale for Your Responses

For each question, you will be comparing two items and deciding which one is more important, and by how much. You will use a scale from 1 to 9, where:

- **1: Equal importance** (Both items are equally important)
- **3: Moderate importance** (The first item is moderately more important than the second)
- **5: Strong importance** (The first item is strongly more important than the second)
- **7: Very strong importance** (The first item is very strongly more important than the second)
- **9: Extreme importance** (The first item is overwhelmingly more important than the second)

You can also use **2, 4, 6, 8** for intermediate values.

What if the SECOND item is more important?
If the second item in the comparison is more important, you should select the **reciprocal value** (e.g., if the second item is moderately more important, select **1/3,** Similarly, if second item is strongly more important than the first one enter **1/7**).

1. **Question 1:** How much more important is **'Material degradation'** compared to **'Foundation/scour vulnerability'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

2. **Question 2:** How much more important is **'Material degradation'** compared to **'Structural deformation'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

3. **Question 3:** How much more important is **'Material degradation'** compared to **'Joint condition'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

4. **Question 4:** How much more important is **'Material degradation'** compared to **'Deck integrity'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

5. **Question 5:** How much more important is **'Material degradation'** compared to **'Live load response'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

6. **Question 6:** How much more important is **'Material degradation'** compared to **'Environmental exposure'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

7. **Question 7:** How much more important is **'Foundation/scour vulnerability'** compared to **'Structural deformation'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

8. **Question 8:** How much more important is **'Foundation/scour vulnerability'** compared to **'Joint condition'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

9. **Question 9:** How much more important is **'Foundation/scour vulnerability'** compared to **'Deck integrity'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

10. **Question 10:** How much more important is **'Foundation/scour vulnerability'** compared to **'Live load response'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

11. **Question 11:** How much more important is **'Foundation/scour vulnerability'** compared to **'Environmental exposure'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

12. **Question 12:** How much more important is **'Structural deformation'** compared to **'Joint condition'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

13. **Question 13:** How much more important is **'Structural deformation'** compared to **'Deck integrity'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

14. **Question 14:** How much more important is **'Structural deformation'** compared to **'Live load response'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

15. **Question 15:** How much more important is **'Structural deformation'** compared to **'Environmental exposure'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

16. **Question 16:** How much more important is **'Joint condition'** compared to **'Deck integrity'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

17. **Question 17:** How much more important is **'Joint condition'** compared to **'Live load response'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

18. **Question 18:** How much more important is **'Joint condition'** compared to **'Environmental exposure'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

19. **Question 19:** How much more important is **'Deck integrity'** compared to **'Live load response'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

20. **Question 20:** How much more important is **'Deck integrity'** compared to **'Environmental exposure'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

21. **Question 21:** How much more important is **'Live load response'** compared to **'Environmental exposure'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

22. **Question 22:** If you think a few **additional indicators** could improve the assessment of the structural condition, **please suggest them** and briefly explain how each can be evaluated (e.g., Joint Condition: Visual check for looseness, corrosion, or missing elements)

SECTION 3: RELATIVE IMPORTANCE OF CULTURAL INDICATORS

In this section, please compare pairs of **cultural indicators** to determine their relative importance in assessing the heritage value of historic bridges.

Cultural Indicators:

- Historic age (e.g., age since original construction, historical significance of era)
- Traditional construction method (e.g., use of local joinery, stone dressing, unique techniques)
- Architectural uniqueness (e.g., distinctive design, proportions, ornamentation)
- Local cultural importance (e.g., ritual use, symbolism, association with festivals or oral history)

Understanding the Scale for Your Responses

For each question, you will be comparing two items and deciding which one is more important, and by how much. You will use a scale from 1 to 9, where:

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- **3: Moderate importance** (The first item is moderately more important than the second)
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- **7: Very strong importance** (The first item is very strongly more important than the second)
- **9: Extreme importance** (The first item is overwhelmingly more important than the second)

You can also use **2, 4, 6, 8** for intermediate values.

What if the SECOND item is more important?
If the second item in the comparison is more important, you should select the **reciprocal value** (e.g., if the second item is moderately more important, select **1/3,** Similarly, if second item is strongly more important than the first one enter **1/7**).

23. **Question 1:** How much more important is **'Historic age'** compared to **'Traditional construction method'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

24. **Question 2:** How much more important is **'Historic age'** compared to **'Architectural uniqueness'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

25. **Question 3:** How much more important is **'Historic age'** compared to **'Local cultural importance'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

26. **Question 4:** How much more important is **'Traditional construction method'** compared to **'Architectural uniqueness'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

27. **Question 5:** How much more important is **'Traditional construction method'** compared to **'Local cultural importance'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

28. **Question 6:** How much more important is **'Architectural uniqueness'** compared to **'Local cultural importance'**? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

29. **Question 7:** If you think a few **additional indicators** could improve the assessment of the heritage value, **please suggest them** and briefly explain how each can be evaluated (e.g., Architectural Uniqueness: Compare with regional typologies)

SECTION 4: COMPARISON BETWEEN STRUCTURAL AND CULTURAL IMPORTANCE

Overall Importance of Structural vs. Cultural AspectsIn this final section, please rate the relative importance of **Structural** considerations versus **Cultural** considerations when making overall decisions about historic bridge conservation.

Understanding the Scale for Your Responses

For each question, you will be comparing two items and deciding which one is more important, and by how much. You will use a scale from 1 to 9, where:

- **1: Equal importance** (Both items are equally important)
- **3: Moderate importance** (The first item is moderately more important than the second)
- **5: Strong importance** (The first item is strongly more important than the second)
- **7: Very strong importance** (The first item is very strongly more important than the second)
- **9: Extreme importance** (The first item is overwhelmingly more important than the second)

You can also use **2, 4, 6, 8** for intermediate values.

What if the SECOND item is more important?
If the second item in the comparison is more important, you should select the **reciprocal value** (e.g., if the second item is moderately more important, select **1/3,** Similarly, if second item is strongly more important than the first one enter **1/7**).

30. **Question 1:** How much more important are **'Structural'** considerations compared to **'Cultural'** considerations? *(Input a number from 1 to 9, or a reciprocal like 1/3 or 1/7)* *

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